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RESEARCH ARTICLE

Temporal Preparation in Athletes: A Comparison of Tennis Players and Swimmers With Sedentary Controls

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ABSTRACT. The authors aimed to investigate the effects of different sporting experience on nonspecific temporal preparation. They evaluated temporal preparation in tennis players (an open-skill sport) and their athletic (swimmers, a closed skill-sport) and nonathletic (sedentary students) controls using a go/no-go variable foreperiod paradigm in which one simple condition and two go/no-go conditions (central-go and mixed-go) were included, which can be used to study the temporal aspects of nonspecific preparation with decision making in inhibition with different levels of cognitive load. Tennis players responded faster than nonathletic controls while there was no significant difference relative to the athletic controls. Additionally, the main finding of the present study is that the difference in reaction time between tennis players and nonathletic controls was found selectively for short foreperiods in which temporal uncertainty is higher and less temporal preparation can occur. Moreover, correlation analysis revealed that superior temporal preparation was positively associated with enhanced go/no-go decision making in the higher difficulty condition. Our findings are consistent with tennis players showing superior temporal processing. The absence of a significant effect in athletic controls suggests that there is a specific benefit from tennis training and indicates that temporal preparation may be susceptible to modulation by fitness and appropriate training.

Keywords: foreperiod effect, reaction time, sports, temporal processing

When required to act quickly upon forthcoming events, individuals usually try to anticipate when the events will occur covertly in order to optimize performance. This speed-related performance facilitation can be attributed to temporal preparation (Nobre, Correa, & Coull, 2007; Woodrow, 1914), a function that is common and essential in daily life. For example, when driving a car and waiting for a traffic light to change individuals tend to prepare the motor actions such that they can rapidly respond to the change of the light from red to green. In sport, especially for open-skill sports that require adjustment of movements in response to changes in the sporting environment, temporal preparation is crucial for successful performance. For instance, Bootsma and van Wieringen (1990) indicated that it is nearly impossible to return a strong approaching drive in table tennis at the speed of 150 km/hr or faster successfully without preparation. One way to achieve success under such time-limited conditions is to use anticipatory cues to reduce response delay times (e.g., Shim, Carlton, Chow, & Chae, 2005). It has been demonstrated that such anticipatory information may orient attention to certain timings, resulting in enhanced temporal preparation (Miniussi, Wilding, Coull, & Nobre,

1999). However, it remains unknown whether this superiority in preparatory processing persists in the absence of an anticipatory cue, namely, nonspecific temporal preparation (Nobre et al., 2007; Woodrow, 1914).

Recent studies have reported beneficial effects of engaging in exercise or competitive sport training on cognitive and brain functioning, which can be observed as both superior performance and more efficient brain networks (Hillman, Erickson, & Kramer, 2008; Yarrow, Brown, & Krakauer, 2009). Similarly, these facilitative effects can also be seen in preparatory processes (e.g.: Hillman, Weiss, Hagberg, & Hatfield, 2002; Kamiyo, O'Leary, Pontifex, Themanson, & Hillman, 2010; Morillo, Di Russo, Pitzalis, & Spinelli, 2006; Renaud, Bherer, & Maquestiaux, 2010). In order to investigate the influence of aerobic fitness on preparatory processes, Hillman et al. (2002) used an electrophysiological index, contingent negative variation (CNV), to explore this difference between aerobically fit and sedentary individuals. The results showed that although no significant behavioral difference was observed, the CNV amplitude for aerobically trained individuals was much smaller compared with sedentary participants. Thus, the difference in CNV amplitude might be related to less neural resources being allocated to task execution for aerobically fit participants, indicating increased efficiency in their response preparation (Hillman et al., 2002).

Although aerobic fitness and physical activity may modulate temporal preparation ability, it has been suggested that the nature of the sport type should be taken into account in these types of studies (Mann, Williams, Ward, & Janelle, 2007; Overney, Blanke, & Herzog, 2008; Voss, Kramer, Basak, Prakash, & Roberts, 2010; Yarrow et al., 2009). Generally, in open-skill sports (e.g., basketball, tennis, badminton), players are required to react in a dynamically changing, unpredictable, and external-directed environment. In contrast, the sporting environment is relatively stable, predictable, and self-directed for players in closed skill sports (e.g., running, swimming; Di Russo et al., 2010). Hence, these differences in the sporting activity environment may exert different pressures and so develop athletes' cognitive skills in different ways. A meta-analysis study has revealed

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that athletes in open-skilled sports such as tennis players responded faster than those in close-skilled sports, such as swimmers (Mann et al., 2007), indicating the importance of an appropriate athletic comparison group (Voss et al., 2010) when examining this type of effect.

Temporal preparation can be investigated using a paradigm in which the foreperiod (FP) between a warning signal (WS) and an imperative signal (IS) are changed randomly from trial to trial (Niemi & Näätänen, 1981; Nobre et al., 2007). In this paradigm, participants ideally maintain an elevated preparatory state because the time uncertainty about the target is high (Klemmer, 1956). Studies employing the variable FP paradigm show that the length of the FP is inversely related to response time (RT), namely, the longer the FP, the shorter the RT (Los & Heslenfeld, 2005; Los, Knol, & Boers, 2001). This FP-based RT difference is termed the FP effect (Niemi & Näätänen, 1981; Woodrow, 1914) and has been construed as an endogenous and strategic process (Niemi & Näätänen, 1981). It is suggested that if the FPs occur with equal probability, the conditional probability of the imperative signal increases progressively as time elapses. Participants are believed to exploit this probability change to strategically increase preparatory processes due to their anticipation about the moment of IS presentation rising with relatively longer FPs, thus resulting in shorter RTs (Los & Heslenfeld, 2005; Niemi & Näätänen, 1981; Nobre et al., 2007). Such dynamic modulation in temporal preparation and anticipation is also termed the hazard functions (Nobre et al., 2007). In addition, because the WS conveys no information about the imperative signal, this preparatory effect is nonspecific (Bertelson & Boons, 1960; Los & Schut, 2008).

Interestingly, experimental studies have demonstrated that temporal preparation has the capacity to influence cognitive performance (Bausenhardt, Rolke, Hackley, & Ulrich, 2006; Rolke & Hoffmann, 2007). Research has found that subjects showed increased ability to discriminate visual stimuli and responded faster when they showed higher temporal preparation, whereas their performance was the opposite when their temporal preparation was relatively lower (Rolke & Hoffmann, 2007). This result led to the suggestion that temporal preparation modulates perceptual sensitivity. Similar findings can also be seen at the motor level (e.g., Tandonnet, Burle, Vidal, & Hasbroucq, 2003). Therefore, these findings suggest that the temporal deployment of attentional resources for enhancement of cognitive processes may depend on some extent of the level of anticipation. Accordingly, here we were also interested in whether the effect of sporting experience on temporal preparation could also influence decision making.

To examine whether nonspecific temporal preparation is affected by sporting experience, a modified variable FP paradigm was used to compare the difference between athletes and nonathletes. Furthermore, considering the importance of a suitable athletic control groups (Voss et al., 2010), athletes from open- and close-skill sports participated in the present study. We chose tennis players as the open-skilled athletic group because they are trained yearly in a sport that

has intense temporal demands. They are therefore highly likely to show effects in the temporal domain as their skill acquisition progresses (Overney et al., 2008) relative to both untrained individuals and those who participate in close-skill sports. Swimmers were selected as the close-skilled group because they have similar levels of sporting activity and aerobic fitness but do not require the spatiotemporal capacity needed by tennis players (beyond the ability to respond to the start signal at the beginning of a race). Moreover, to determine whether other cognitive processes, such as decision making, would be modulated by the levels of temporal preparation, a simple-go condition with variable FPs and two go/no-go conditions with variable FPs were adopted. A classical FP effect has previously been found in a simple go/no-go paradigm with variable FPs (Cravo, Rohenkohl, Wyart, & Nobre, 2011).

We hypothesized that (a) athletes would have better temporal preparation than nonathletes due to difference in levels of physical activity and aerobic fitness (Hillman et al., 2002; Kamiyo et al., 2010; Renaud et al., 2010); (b) given the importance of sport type (Voss et al., 2010), temporal preparation would be modulated more in athletes from an open-skill sport relative to those from a closed-skill sport; (c) participants who show superior ability in temporal preparation would show better decision making because it is suggested that temporal preparation has the capacity to modulate cognitive processes (Rolke & Hoffmann, 2007; Tandonnet et al., 2003). To the best of our knowledge, the present study is the first to examine the effect of engaging in sport on temporal preparation in young adults using a variable FP paradigm and, more importantly, the first study assessing differences in temporal preparation between athletes engaging in different categories of sport type.

Method

Participants

Forty-two male university students participated in the present study. Of these, 14 students were members of the university tennis team (M age = 20.64 years, SD = 2.76 years; tennis experience = 3–11 years), while another 14 students belonged to the university swim team (M age = 19.36 years, SD = 0.74 years; swim experience = 2.5–9 years). Additionally, all of the athletes who participated in the study were qualified for the National Intercollegiate Athletic Games in Taiwan. The remaining fourteen students reported having no specialty in any sport/exercise and were sedentary (M age = 21.21 years, SD = 2.61 years). Table 1 summarizes the participants' demographic information. All participants had normal or corrected-to-normal visual acuity and were right-handed. No individuals reported having a history of neurological problems or cardiovascular diseases, nor were any taking any medications that affect cognitive functions. All participants provided informed consent prior to participating and the study was approved by the local ethics committee.

TABLE 1. Demographics of Participants in Each Group

Group	Tennis players (<i>n</i> = 14)		Swimmers (<i>n</i> = 14)		Sedentary controls (<i>n</i> = 14)	
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>
Age (years)	20.64	2.76	19.36	0.74	21.21	2.61
Height (m)	1.75	0.06	1.72	0.06	1.73	0.05
Weight (kg)	70.64	10.97	67.62	5.09	63.96	8.68
BMI (kg/m ²)	22.97	2.38	22.95	2.56	21.50	3.29
Kilocalorie expenditure (Kcal/day)	2694.07	455.82***	2853.78	348.62***	2074.59	303.55
VO ₂ max (ml*kg ⁻¹ *min ⁻¹)	55.32	2.69***	56.32	2.48***	44.52	2.59

****p* < .001.

Measures

Seven-Day Physical Activity Recall Questionnaire

A seven-day physical activity recall questionnaire was adapted from Sallis et al. (1986), which can be utilized to estimate and quantify the participants' level of physical activity (Maximova, O'Loughlin, Paradis, Hanley, & Lynch, 2009). The seven-day physical activity recall questionnaire allowed the application of relatively strict screening criteria to help ensure that the sporting groups did not differ in their levels of physical activity and so potentially bias the results. Thus, participants were excluded according to the following criteria: (a) for the athletic groups, those who engaged in their training programs less than three times per week; and (b) for the control group, those who spent more than 1 hr per week exercising at the intensity of moderate or higher.

Aerobic Fitness Evaluation

Previously studies had suggested that temporal preparation might be associated with fitness level (Hillman et al., 2002; Renaud et al., 2010). In order to take this into account the present study adopted a nonexercise formula (Bradshaw et al., 2007; Chan, Wong, Liu, Yu, & Yan, 2011) to evaluate the participants' VO₂max, an index of aerobic fitness. This formula has the capacity to provide an appropriate prediction of the actual VO₂max (*R* = .93) in adults 18–65 years old (Bradshaw et al., 2007). In this evaluation, descriptive characteristics for participants' (age, height, weight, gender, and body mass index [BMI]) were required. In addition, the Physical Activity Rating (PA-R; ranging from 0 to 10 points; Bradshaw et al., 2007), and the Perceived Functional Ability (PFA; ranging from 2 to 26 points; George, Stone, & Burkett, 1997) questionnaires were used to survey the self-reported capacity to walk, jog, or run between one and three miles (PFA), and physical activity level for the previous six months (PA-R). After collecting the required information, the levels of aerobic fitness were quantified by the formula (Bradshaw et al., 2007; Chan et al., 2011): VO₂max (mlkg⁻¹min⁻¹) = 48.073 +

$$(6.178 \times \text{Gender; female} = 0, \text{male} = 1) - (0.246 \times \text{Age}) - (0.619 \times \text{BMI}) + (0.712 \times \text{PFA}) + (0.671 \times \text{PA-R}).$$

Go/No-Go Variable FP Paradigm

The present study employed a go/no-go paradigm with variable FPs, consisting of one simple RT condition and two go/no-go conditions. This paradigm allows decision making in inhibition can be investigated and was adopted from Nakamoto and Mori (2008)'s study. The key parameter we modified is that we changed the FP design from a fixed design to a variable design has this allows examination of nonspecific temporal preparation (Niemi & Näätänen, 1981).

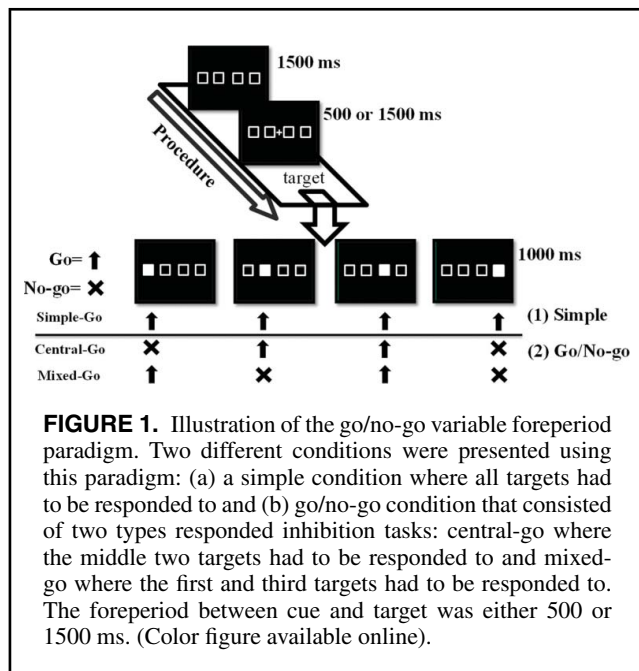
The paradigm was programmed in E-prime (Psychology Software Tools, Inc, Sharpsburg, PA). All stimuli were presented using a 17-inch cathode ray tube (CRT) screen using a video resolution of 1024 × 768 pixels and a vertical refresh rate of 100 Hz. Similar to Nakamoto and Mori (2008)'s study, the possible locations of four-square frames (each approximately 3.4° × 3.4°) were equally spaced along a horizontal row of 27.4° on the screen. The screen was presented in black, and a target was presented in green. The procedure of the task is illustrated in Figure 1. The warning signal was presented as a white fixation point, and was followed by the target after a random FP of either 500 ms (FP500) or 1500 ms (FP1500) with equal probability, which is the time range commonly used in studies investigating the mechanisms of temporal preparation (Niemi & Näätänen, 1981; Vallesi, Shallice, & Walsh, 2007). The intertrial interval (ITI) was 1500 ms. The following sections described the conditions of the paradigm.

Simple Condition

The participants were instructed to respond to the target as quickly as possible, regardless of the position it was presented in.

Go/No-Go Conditions

There were two conditions in the response inhibition task: the central-go and the mixed-go condition, which may help



evaluate participants' decision making with different levels of difficulty. In the central-go condition, participants were instructed to respond to either of the two central targets when they were presented as quickly and accurately as possible. When the target was presented in one of the two peripheral locations their response had to be inhibited. On the other hand, during the mixed-go condition, participants had to execute or inhibit their response based on the assigned position of the target (e.g., from left to right the appropriate responses could be go, no-go, go, no-go). Two kinds of response mapping sequence were counterbalanced among the participants (see Figure 1).

Participants were instructed to respond to the target by pressing a button on a response box with the index finger of their dominant hand as fast as possible when a Go-target appeared and to refrain from responding to No-go targets. The order of the three conditions was counterbalanced across all participants.

Procedure

All participants took part in the experimental procedure that included two phases: the first phase consisted of the seven-day physical activity recall questionnaire and fitness evaluation, the second phase was comprised of the cognitive task. Only eligible participants who pass the criterion for the questionnaire went on to perform the cognitive task. The duration of the total experimental procedure was approximately 30 min.

Data Analysis

All participants' descriptive characteristics (including age, height, weight, BMI, physical activity level, and aerobic fit-

ness level) were described using means and standard deviations, and group differences were analyzed using a one-way analysis of variance (ANOVA) method. A three-way 3 Group (tennis, swim, control) $\times$ 3 Condition (simple, central-go, mixed-go) $\times$ 2 FP (500 ms, 1500 ms) mixed design, factorial, and repeated measures ANOVA with a Bonferroni adjustment for multiple comparisons was conducted to analyze the error rate and RTs for correct go trials. Two types of responses were considered as commission errors and were excluded from analysis: (a) failure to inhibit responses in no-go trials; (b) RTs for go trials beyond the range of the mean $\pm$ 2 standard deviations. Group was as a between-participants factor and the condition and FP were within-participants factors. In addition, in order to examine the difference in the magnitude of the FP effect across groups, this was transformed into a normalized value by the following equation: $(RT_{FP500ms} - RT_{FP1500ms})/RT_{FP500ms}$. The normalized FP effect was submitted to a one-way ANOVA with a Bonferroni adjustment to examine the group effect. Finally, we also examined whether the other cognitive processes, such as decision making, were affected by temporal preparation. Thus, RT differences from simple-go and go/no-go conditions were calculated as the magnitude of decision making (for central-go condition: $Cen500 - Simple500$; for mixed-go condition: $Mix500 - Simple500$). A bivariate analysis (Pearson correlation) was then conducted to examine the correlation between the magnitude of decision making and the FP effect, to investigate whether participants with better temporal preparation showed enhanced decision making. The level of significance was set at $p \leq .05$. All analysis was completed with the SPSS version 20.0 software system.

Results

Participant Demographics

Participant demographic data and their levels of physical activity are provided in Table 1. Demographic variables including age, $F(2, 39) = 2.54, p = .092, \eta_p^2 = .12$; height, $F(2, 39) = 1.00, p = .376, \eta_p^2 = .05$; weight, $F(2, 39) = 2.12, p = .134, \eta_p^2 = .01$; and BMI, $F(2, 39) = 1.35, p = .270, \eta_p^2 = .07$, did not differ between groups. Furthermore, there was a significant group difference for the levels of physical activity, $F(2, 39) = 16.90, p < .001, \eta_p^2 = .46$, and aerobic fitness, $F(2, 39) = 89.41, p < .001, \eta_p^2 = .82$. Post hoc pairwise comparisons showed that tennis players and swimmers had similar level of physical activity, $t(26) = -1.13, p = .799, d = 0.39$, and aerobic fitness, $t(26) = 1.02, p = .939, d = -0.039$, whereas both the athletic groups had higher levels of physical activity and aerobic fitness when compared to the sedentary controls: physical activity, tennis players versus sedentary controls, $t(26) = 4.38, p < .001, d = 1.60$ / aerobic fitness, $t(26) = 1.02, p < .001, d = 4.09$; swimmers versus sedentary controls, physical activity, $t(26) = 5.50, p < .001, d = 2.38$ / aerobic fitness, $t(26) = -0.11, p < .001, d = 4.65$.

TABLE 2. Error Rate Performances of Participants in Each Group

FPs	Tennis players (<i>n</i> = 14)				Swimmers (<i>n</i> = 14)				Sedentary controls (<i>n</i> = 14)			
	500ms		1500 ms		500 ms		1500 ms		500 ms		1500 ms	
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>
Simple-go (%)	—	—	—	—	—	—	—	—	—	—	—	—
Central-go (%)	2.8	3.1	1.8	2.7	2.3	2.7	0.8	1.5	0.3	0.1	2.8	2.9
Mixed-go (%)	2.0	4.1	3.7	3.8	2.6	2.3	2.4	3.1	2.0	2.7	0.7	1.7

Commission Error

For the commission error analysis, we only compared the central-go and mixed-go conditions because there were no errors in the simple condition due to the nature of the task (see Table 2). The data showed no main effect for condition, $F(2, 39) = 1.95$, $p = .17$, $\eta_p^2 = .04$; FP, $F(2, 39) = 0.00$, $p = .97$, $\eta_p^2 = .00$; or group, $F(2, 39) = 1.62$, $p = .97$, $\eta_p^2 = .21$. However, a condition by FP by group interaction was observed, $F(2, 39) = 1.62$, $p = .01$, $\eta_p^2 = .23$.

RT

Figure 2a shows the mean RT across groups and conditions. A significant main effect was observed across

conditions, $F(2, 39) = 227.99$, $p < .001$, $\eta_p^2 = .85$, with shortest RTs for the simple condition ($M = 300.25$ ms, $SD = 25.82$ ms), intermediate RTs for the central-go condition ($M = 368.04$ ms, $SD = 30.94$ ms), and slowest RTs for the mixed-go condition ($M = 396.72$ ms, $SD = 35.32$ ms). We also found a main effect for FP, $F(2, 39) = 28.68$, $p < .001$, $\eta_p^2 = .42$. RTs for the shorter FP ($M = 362.88$ ms, $SD = 30.01$ ms) were much slower compared with the longer FP ($M = 347.13$ ms, $SD = 25.12$ ms), showing the classical FP effect in go/no-go conditions (Cravo et al., 2011). Moreover, a significant effect for group was observed, $F(2, 39) = 3.39$, $p = .044$, $\eta_p^2 = .15$, and post hoc analysis showed that tennis players ($M = 345.21$ ms, $SD = 23.14$ ms) had faster RTs than sedentary controls ($t(26) = -2.52$, $p = .048$, $d = -1.00$),

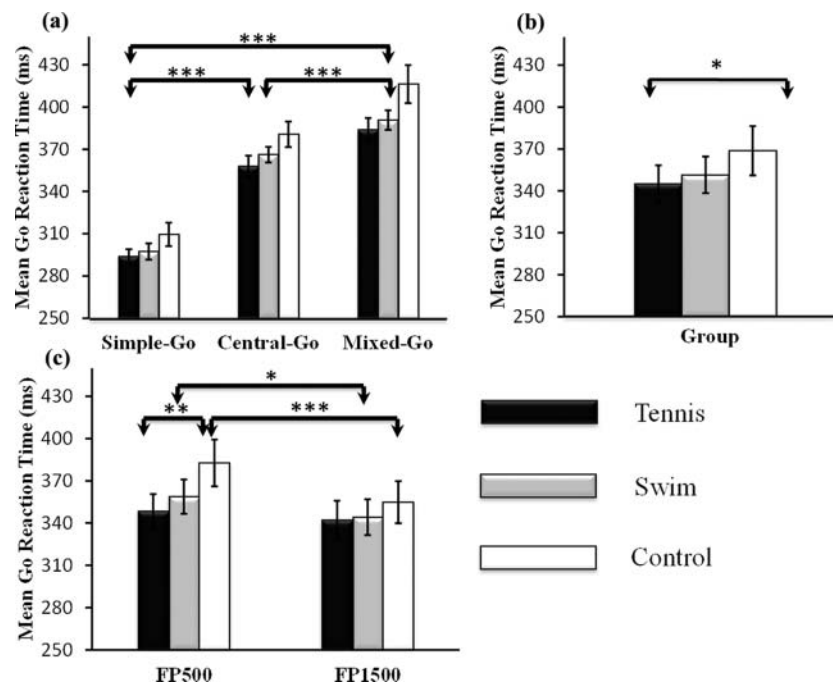


FIGURE 2. Mean of the go reaction times and standard error (in ms) for each analysis. (a) Mean go reaction times for simple-go, central-go, and mixed-go conditions ($***p < .001$). (b) Mean go reaction times for tennis players, swimmers, and sedentary controls ($*p < .05$). (c) Mean go reaction times for each FP500 ms and FP1500 ms across groups ($*p < .05$. $**p < .01$. $***p < .001$).

while no RT difference was found for other groups comparisons (tennis players: $M = 345.21$ ms, $SD = 23.14$ ms vs. swimmers, $M = 351.45$ ms, $SD = 16.48$ ms; swimmers vs. controls: $M = 368.36$ ms, $SD = 31.15$ ms; see Figure 2b). Critically, a FP by group interaction was present, $F(2, 39) = 4.02$, $p = .026$, $\eta_p^2 = .17$, the classical FP effect (decreasing RTs with increasing FP) was present in both swimmers and sedentary controls but absent in the tennis players (Figure 2c). Finally, a one-way ANOVA with Bonferroni-corrected t tests showed that RT for tennis players ($M = 348.30$ ms, $SD = 23.86$ ms) was faster than sedentary controls ($M = 381.62$ ms, $SD = 36.70$ ms) selectively for the shorter FP, $t(26) = -3.15$, $p = .009$, $d = -1.24$ (Figure 2c).

Normalized FP Effect

The main effect of the group factor was significant, $F(2, 39) = 3.68$, $p = .034$, $\eta_p^2 = .116$. Results of post hoc analysis showed that tennis players ($M = 0.017$ ms, $SD = 0.04$ ms) had a significantly smaller normalized FP effect in comparison with sedentary controls ($M = 0.067$ ms, $SD = 0.05$ ms), $t(26) = -2.71$, $p = .030$, $d = -1.06$, whereas no significant difference was found between swimmers ($M = 0.040$ ms, $SD = 0.05$ ms) and sedentary controls, $t(26) = -1.48$, $p = .437$, $d = -0.58$, or between tennis players and swimmers, $t(26) = -1.23$, $p = .683$, $d = -0.48$ (Figure 3).

The Effect of Temporal Preparation on Decision Making

In this analysis we aimed to examine whether expertise in tennis benefited other cognitive processes, such as decision making, on top of temporal preparation. Because the central-GO and mixed-go conditions require one extra stage of cognitive processing, namely discrimination of the stimuli and a subsequent decision, on top of simply pressing a button, any benefit in addition to temporal preparation should be reflected by a greater magnitude of facilitation in the mixed-go

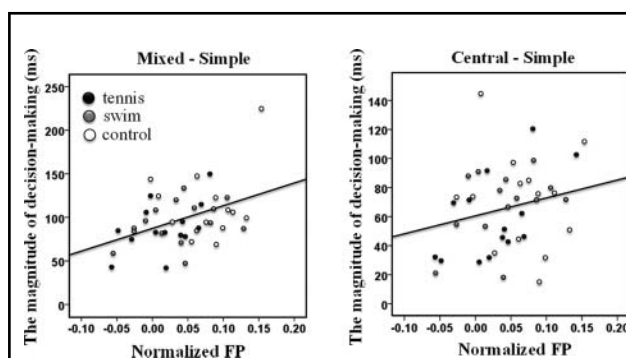


FIGURE 4. The correlation between the magnitudes of foreperiod effect and decision-making among participants. The normalized foreperiod (FP) denotes the time cost induced by temporal uncertainty. The magnitude of decision making denotes the time cost induced by decision making in response inhibition. (a) The correlation between the magnitude of FP effect and decision making in mixed-go condition; (b) the correlation between the magnitude of FP effect and decision making in central-go condition.

or central-go condition. To this end, we subtracted $RT_{\text{simple-go}}$ from $RT_{\text{central-go}}$ and $RT_{\text{mixed-go}}$ for the FP500 to test the magnitude of facilitation for each go/no-go condition, and to examine its association with normalized FP effect, an index of time cost induced by temporal uncertainty. We found that the magnitude of FP effect was positively correlated with decision making in the mixed-go condition ($r = .416$, $p = .006$; Figure 4a), whereas no such effect was found in the central-go condition ($r = .226$, $p = .150$; Figure 4b).

Discussion

The purpose of the present study was to determine whether there are differences in temporal preparation with no pre-information (e.g., anticipatory cue) across athletes in open-skilled sports, athletes in close-skilled sports and nonathletes. First, we hypothesized that both athletic groups would exhibit better temporal preparation than the nonathletic controls due to the difference in physical fitness between athletes and nonathletes (Hillman et al., 2002; Kamijo et al., 2010). Second, owing to that fact that tennis players are trained under strong temporal demands and have superior ability in temporal processing (Overney et al., 2008), we expected that temporal preparation might be more affected in tennis players relative to swimmers, which means better temporal preparation would be predicted in the tennis players. Last, we also hypothesized that those who have superior ability in temporal preparation might also show better performance in other cognitive processes such as decision making. Accordingly, we adopted a variable FP go/no-go paradigm to test these three hypotheses.

The key finding of this study is that tennis players responded faster on the task than sedentary controls. Specifically, this difference was found selectively in the shorter FP

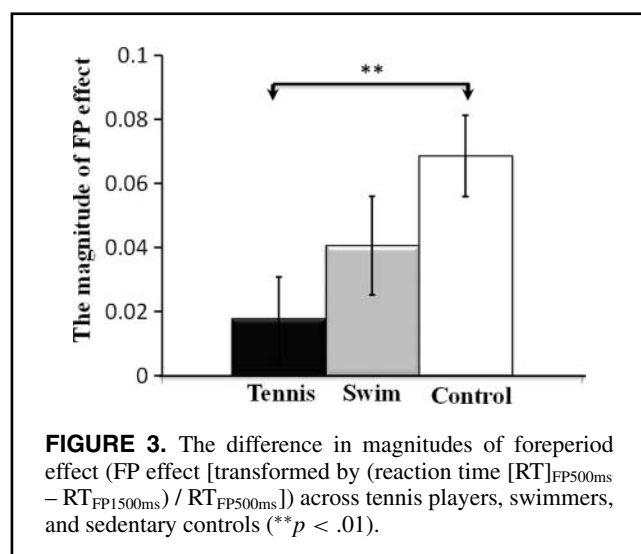


FIGURE 3. The difference in magnitudes of foreperiod effect (FP effect [transformed by (reaction time [RT]_{FP500ms} - RT_{FP1500ms}) / RT_{FP500ms}]) across tennis players, swimmers, and sedentary controls (** $p < .01$).

trials. Moreover, we evaluated the magnitude of the FP effect by using a transformed value and found that tennis players had significant smaller FP effects than the sedentary controls, suggesting the temporal uncertainty is lower in tennis players. The pattern of effects for swimmers, who did not differ from the tennis players but, importantly, did also not differ from sedentary controls, suggests that their performance lies between the two other groups. This is consistent with there being a benefit of exercise in general, as in terms of levels of physical activity and fitness the swimmers and tennis players did not differ, but the additional benefits conferred by tennis specific training were required to cause a significant difference from sedentary subjects. Finally, we found a positive correlation between the magnitude of FP effect and decision making only in the condition (mixed-go) which had a higher difficulty (indicated by longer RT) for participants, suggesting that those who were less affected by temporal uncertainty may perform better in go/no-go decision making when there is a higher difficulty.

The fact that the data show that only tennis players responded significantly faster than sedentary controls, while no difference was found between swimmers and sedentary controls, indicates the importance of athletic comparison groups in this type of study (Mann et al., 2007; Voss et al., 2010). These findings are partially consistent with previous studies with different cognitive tasks examining response preparation (Hillman et al., 2002; Kamijo et al., 2010), which found no significant behavioral differences between aerobically trained groups and sedentary controls. No difference between swimmers and sedentary controls may occur because, despite the beneficial effect of aerobic fitness on cognition (Hillman et al., 2008), sport-related cognitive skills might play an important role in this type of task. That is, training-induced neural changes supporting the athletes' achievement may be necessary for differences to be seen (Miller & Clapp, 2011; Yarrow et al., 2009). The absence of a difference between tennis players and swimmers may be due to their similar level of sporting activity and aerobic fitness, which has been demonstrated to be positively associated with temporal preparation (Hillman et al., 2002; Kamijo et al., 2010; Renaud et al., 2010). It is also possible that there is a small element of temporal preparation training for the swimmers as they are required to learn to react as quickly as possible to a starting signal. Thus, both skill and aerobic training might benefit temporal preparation in the present study, with the combination producing a larger effect than each individually (Chan et al., 2011).

On the other hand, in addition to the sport type, the nature of cognitive tasks used in investigations is clearly of importance as well. Temporal preparation can be readily studied using the variable FP paradigm (Los & Schut, 2008; Niemi & Näätänen, 1981; Vallesi et al., 2007). In such a paradigm, if the FPs are embedded randomly and unexpectedly with equal probability between trials, temporal preparation would be modulated according to the changing of conditional probability, producing the FP effect (i.e., reduced RTs as the FP

increases; Niemi & Näätänen, 1981). The tasks employed in our study were designed with two FPs: 500 ms and 1500 ms, thus the conditional probability of each FP should be 0.5 for FP500 and 1 for FP1500. Consequently, the level of temporal preparation would be lower in the FP500 because of the higher temporal uncertainty, and vice versa. Our results showed that the tennis players had similar RT for both FPs while the other groups responded slower for the shorter FP, which suggests that swimmers and sedentary controls might experience higher temporal uncertainty for the shorter FP, whereas tennis players might maintain a more constant level of temporal preparation irrespective of the time uncertainty. This is supported by the transformed value of the FP effect reflecting the difference in temporal uncertainty for the two FPs, which exhibited a smaller magnitude of FP effect in tennis players when compared to the sedentary controls. This effect however was smaller and not significant when compared to the FP effect in swimmers.

In sum, these findings imply that the level of temporal preparation for both FPs may be relatively equivalent in tennis players as compared to the other groups. Crucially, the major effects of group difference are selective for the shorter FP. This finding is similar to Renaud et al.'s (2010) study showing that high-fitness elderly adults had faster RTs in the shortest FP than their low-fitness counterparts during a short temporal window design (1 s, 3 s, 5 s). Therefore, our data may indicate that the ability to synchronize temporal preparation with the moment of target presentation is superior in tennis players or they had rapid temporal preparation. These results imply that engaging in tennis training may lead to facilitation in temporal preparation, further supporting the notion that playing tennis improves cognitive processing in the temporal domain (Overney et al., 2008). In fact, Overney et al., using basic visual tasks, observed that tennis players had superior capacity in time-related visual perception, and visual perceptual processing can be enhanced through suitable temporal preparation (Rolke & Hoffmann, 2007). These findings lead us to further suggest that playing open-skill sports such as racket sports may improve individuals in temporal preparation without anticipatory information. Data collected here are in support of this argument. Further research examining athletes with different expertise levels and in different open-skill sports would be beneficial in providing further insight into this possibility. This would also allow assessment of whether it is tennis training that results in alterations in temporal preparation ability or whether such an initial ability is associated with better performance at the sport.

To explore whether other cognitive processes would be facilitated in association with superior temporal preparation, we adopted a go/no-go paradigm with variable FPs, allowing the evaluation in decision making in inhibition during different levels of temporal preparation. The correlation analysis revealed the positive relationship between the magnitude of FP effect and decision making, which is in line with previous studies showing that better temporal preparation is associated

with enhanced cognitive processing (e.g., perceptual processing or motor processing; Rolke & Hoffmann, 2007; Tandonnet et al., 2003). This result also suggests that individuals with superior temporal preparation may show enhanced decision making in terms of inhibition relative to those with inferior temporal preparation. In the present study, tennis players showed superiority in temporal preparation, which may further facilitate their decision making in response inhibition, when compared to sedentary controls. However, this effect didn't occur in the condition with lower difficulty. The absence of this correlation may be due to, for example, the central-go condition used not being difficult enough to reveal the facilitative effect of temporal preparation on decision making. Future researchers should explore the interaction between levels of temporal preparation and task difficulty.

The present study suggests that tennis players may have a superior ability to temporally prepare for an upcoming event, which may be due their greater ability to synchronize the moments between temporal preparation and target presentation. This may be a consequence of tennis training under conditions in which temporal information is always under high uncertainty. This could be investigated by future researchers examining whether temporal preparation differs with different levels of tennis training. Although aerobic training has been indicated to be positively associated with response preparation, this beneficial effect was not seen in our study. Thus, it is of interest to investigate the characteristics of temporal preparation for athletes in different categories of sports.

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